ChatGPT

give me basic questions on string+dsa for interview purposes'

Here are some basic interview questions on strings and Data Structures & Algorithms (DSA) related to string manipulation:

1. Reverse a String

- Write a function to reverse a given string without using the built-in reverse function.
- Input: `"hello"`
- Output: `"olleh"`

2. Check for Palindrome

- Given a string, write a function to check if it is a palindrome (i.e., the string reads the same forward and backward).
- Input: `"madam"`
- Output: `True`

3. Count Occurrences of a Character

- Write a function to count the occurrences of a character in a string.
- Input: `"banana", 'a'`
- Output: `3`

4. Anagram Check

- Write a function to check if two strings are anagrams of each other (contain the same characters in the same frequency).
- Input: `"listen", "silent"`
- Output: `True`

5. Longest Prefix

- Write a function that takes an array of strings and finds the longest common prefix among them.
- Input: `["flower", "flow", "flight"]`
- Output: `"fl"`

6. First Non-Repeated Character

- Write a function to find the first non-repeated character in a string.
- Input: `"swiss"`
- Output: `"w"`

7. String Compression

- Given a string, implement basic string compression using the counts of repeated characters. For example, `` `"aabcccccaaa"` `` becomes `` `"a2b1c5a3"` ``.
- Input: `` `"aabcccccaaa"` ``
- Output: `` `"a2b1c5a3"` ``

8. String Permutations

- Write a function that generates all the permutations of a given string.
- Input: `` `"abc"` ``
- Output: `` `["abc", "acb", "bac", "bca", "cab", "cba"]` ``

9. Find the Length of the Longest Substring Without Repeating Characters

- Given a string, find the length of the longest substring that does not have any repeating characters.
- Input: `` `"abcabcbb"` ``
- Output: `` `3` `` (The substring is `` `"abc"` ``)

10. Count and Say Sequence

- The count and say sequence is a sequence of digit strings defined by the recursive formula:
  - `` `countAndSay(1) = "1"` ``
  - `` `countAndSay(n)` `` is the result of reading the digits of `` `countAndSay(n-1)` `` from left to right, grouping the same digits together.
- Given `` `n` ``, generate the nth term of the sequence.
- Input: `` `4` ``
- Output: `` `"1211"` ``

11. String Rotation

- Write a function to check if one string is a rotation of another.
- Input: `` `"waterbottle", "erbottlewat"` ``
- Output: `` `True` ``

12. Valid Parentheses

- Given a string containing just the characters `` `(` ``, `` `)` ``, `` `{` ``, `` `}` ``, `` `[` `` and `` `]` ``, determine if the input string is valid.
- Input: `` `"()[]{}"` ``
- Output: `` `True` ``

13. Remove Duplicate Characters

- Write a function to remove duplicate characters from a string, keeping only the first occurrence of each character.
- Input: `` `"aabbcc"` ``
- Output: `` `"abc"` ``

14. Substring with Concatenation of All Words

- You are given a string `s` and a list of words `words`. Find all starting indices of substring(s) in `s` that is a concatenation of each word in `words` exactly once, without any intervening characters.
- Input: `"barfoothefoobarman", ["foo", "bar"]`
- Output: `[0, 9]`

15. Check if a String Contains All Unique Characters

- Write a function that checks whether a string has all unique characters.
- Input: `"abcdef"`
- Output: `True`

16. Longest Repeating Character Replacement

- Given a string and an integer `k`, you can replace at most `k` characters in the string to make it a string with the longest possible substring of repeated characters.
- Input: `"AABABBA", 1`
- Output: `4` (The longest substring is `"AAAA"`)

17. Longest Palindromic Substring

- Given a string `s`, find the longest palindromic substring in `s`.
- Input: `"babad"`
- Output: `"bab"` (or `"aba"`)

18. Group Anagrams

- Given a list of strings, group the anagrams together.
- Input: `["eat", "tea", "tan", "ate", "nat", "bat"]`
- Output: `[["eat", "tea", "ate"], ["tan", "nat"], ["bat"]]`

19. Find All Occurrences of a Substring

- Write a function to find all occurrences of a substring in a string.
- Input: `"ABABABAB", "AB"`
- Output: `[0, 2, 4, 6]`

20. Replace Spaces with %20

- Write a function to replace all spaces in a string with `%20`.
- Input: `"Mr John Smith"`
- Output: `"Mr%20John%20Smith"`

These questions cover a broad range of fundamental string manipulation concepts and are commonly asked in coding interviews.

or which uses different methods of string class

Here are some interview questions that focus on using different methods of the `String` class in various programming languages (such as Java or Python). These questions explore methods like concatenation, substring extraction, character search, case conversion, splitting, trimming, etc.

1. Check if a String Contains a Substring

- **Question**: Given a string, write a function to check if it contains another string as a substring.
- **Method(s) used**: `contains()`, `indexOf()` (Java), `in` operator (Python).
- Input: `"Hello, World!"`, `"World"`
- Output: `True`

2. Convert a String to Uppercase or Lowercase

- **Question**: Write a function to convert a string to uppercase or lowercase.
- **Method(s) used**: `toUpperCase()`, `toLowerCase()` (Java), `.upper()` and `.lower()` (Python).
- Input: `"hello"`
- Output: `"HELLO"`

3. Trim Leading and Trailing Whitespaces

- **Question**: Write a function to remove leading and trailing spaces from a string.
- **Method(s) used**: `trim()` (Java), `.strip()` (Python).
- Input: `"  Hello, World!  "`
- Output: `"Hello, World!"`

4. Find the First Occurrence of a Character

- **Question**: Write a function to find the first occurrence of a specific character in a string.
- **Method(s) used**: `indexOf()`, `lastIndexOf()` (Java), `.find()` (Python).
- Input: `"Hello, World!"`, `'o'`
- Output: `4` (The first occurrence of `'o'`)

5. Substring Extraction

- **Question**: Write a function to extract a substring from a string.
- **Method(s) used**: `substring()`, `subSequence()` (Java), `.slice()`, `.substring()` (Python).
- Input: `"Hello, World!"`, `7, 12`
- Output: `"World"`

6. Replace All Occurrences of a Substring

- **Question**: Write a function to replace all occurrences of a substring within a string.
- **Method(s) used**: `replace()`, `replaceAll()` (Java), `.replace()` (Python).
- Input: `"banana"`, `"a"`, `"o"`
- Output: `"bonono"`

7. Split a String into Words

- **Question**: Write a function that splits a string by spaces or another delimiter and returns the individual words.
- **Method(s) used**: `split()` (Java), `.split()` (Python).
- Input: `"The quick brown fox"`

- Output: `["The", "quick", "brown", "fox"]`

8. Check if Two Strings Are Equal

- **Question**: Write a function to check if two strings are exactly the same.
- **Method(s) used**: `equals()`, `equalsIgnoreCase()` (Java), `==`, `.equal()` (Python).
- Input: `"hello"`, `"Hello"`
- Output: `False`

9. Count the Number of Occurrences of a Character

- **Question**: Write a function to count how many times a character appears in a string.
- **Method(s) used**: `charAt()`, `indexOf()`, `lastIndexOf()` (Java), `.count()` (Python).
- Input: `"apple"`, `'p'`
- Output: `2`

10. Check if a String Starts or Ends with a Specific Substring

- **Question**: Write a function to check if a string starts or ends with a particular substring.
- **Method(s) used**: `startsWith()`, `endsWith()` (Java), `.startswith()`, `.endswith()` (Python).
- Input: `"Hello, World!"`, `"Hello"`
- Output: `True`

11. Concatenate Two Strings

- **Question**: Write a function to concatenate two strings.
- **Method(s) used**: `concat()` (Java), `+` operator, `.join()` (Python).
- Input: `"Hello"`, `"World"`
- Output: `"HelloWorld"`

12. Check if a String Contains Only Digits

- **Question**: Write a function to check if a string contains only numeric characters.
- **Method(s) used**: `matches()` (Java), `.isdigit()` (Python).
- Input: `"12345"`
- Output: `True`

13. Convert a String to a Character Array

- **Question**: Write a function that converts a string into a character array.
- **Method(s) used**: `toCharArray()` (Java), `list()` (Python).
- Input: `"hello"`
- Output: `['h', 'e', 'l', 'l', 'o']`

14. Reverse a String

- **Question**: Write a function to reverse a given string.
- **Method(s) used**: `reverse()` (Java StringBuilder), slicing (`[::-1]`) (Python).
- Input: `"Hello"`
- Output: `"olleH"`

15. Check if a String is Empty

- **Question**: Write a function to check if a string is empty or null.
- **Method(s) used**: `isEmpty()`, `length()` (Java), `== ""` or `not` (Python).
- Input: `" "`
- Output: `False`

16. Compare Two Strings Ignoring Case

- **Question**: Write a function to compare two strings ignoring case.
- **Method(s) used**: `equalsIgnoreCase()` (Java), `.lower()` or `.upper()` (Python).
- Input: `"hello"`, `"HELLO"`
- Output: `True`

17. Join an Array of Strings into a Single String

- **Question**: Write a function to join an array of strings into a single string with a separator.
- **Method(s) used**: `join()` (Java), `.join()` (Python).
- Input: `["apple", "banana", "cherry"]`, `", "`
- Output: `"apple, banana, cherry"`

18. Find the Last Index of a Character

- **Question**: Write a function to find the last occurrence of a character in a string.
- **Method(s) used**: `lastIndexOf()` (Java), `.rfind()` (Python).
- Input: `"apple"`, `'p'`
- Output: `2`

19. Find the Index of a Substring

- **Question**: Write a function to find the index of the first occurrence of a substring within a string.
- **Method(s) used**: `indexOf()` (Java), `.find()` (Python).
- Input: `"apple"`, `"pl"`
- Output: `1`

20. Replace a Character at a Specific Index

- **Question**: Write a function to replace the character at a specific index in a string.
- **Method(s) used**: Use string slicing and concatenation in Python, or `StringBuilder` in Java.
- Input: `"apple"`, `2`, `'z'`
- Output: `"azple"`

These questions focus on understanding and using various methods available in the string class to perform common operations. They also test how well candidates are familiar with handling strings in different programming languages.

some more questions which will be solved using stringbuilder or buffer

Here are some interview questions that focus on solving problems using `StringBuilder` (or `StringBuffer` in Java), which are more efficient for string manipulation in terms of time complexity compared to using immutable `String` objects:

1. Reverse a String Using StringBuilder

- **Question**: Write a function to reverse a given string using `StringBuilder` or `StringBuffer`.
- **Method(s) used**: `reverse()`
- Input: `"Hello"`
- Output: `"olleH"`

2. Remove All Whitespace from a String

- **Question**: Write a function that removes all whitespaces from a string using `StringBuilder`.
- **Method(s) used**: `StringBuilder.append()`, `charAt()`
- Input: `" H e llo Wo rld "`
- Output: `"HelloWorld"`

3. String Concatenation

- **Question**: Given a list of strings, concatenate them into a single string efficiently using `StringBuilder`.
- **Method(s) used**: `StringBuilder.append()`
- Input: `["Hello", " ", "World"]`
- Output: `"Hello World"`

4. Check if a String is Palindrome Using StringBuilder

- **Question**: Write a function to check if a string is a palindrome by reversing it using `StringBuilder` and comparing it to the original string.
- **Method(s) used**: `StringBuilder.reverse()`, `StringBuilder.toString()`
- Input: `"madam"`
- Output: `True`

5. Remove Duplicates from a String

- **Question**: Write a function to remove duplicate characters from a string using `StringBuilder`.
- **Method(s) used**: `StringBuilder.append()`, `Set` (for tracking seen characters)
- Input: `"aabbcc"`
- Output: `"abc"`

6. Build a String in Reverse Order

- **Question**: Write a function to build a string in reverse order, one character at a time, using `StringBuilder`.
- **Method(s) used**: `StringBuilder.append()`
- Input: `"abcd"`
- Output: `"dcba"`

7. Insert a Character at a Specific Index

- **Question**: Write a function to insert a character at a specified index in a string using `StringBuilder`.
- **Method(s) used**: `StringBuilder.insert()`
- Input: `"abcd"`, index `2`, character `'X'`
- Output: `"abXcd"`

8. Replace All Occurrences of a Substring

- **Question**: Write a function to replace all occurrences of a substring in a string using `StringBuilder`.
- **Method(s) used**: `StringBuilder.indexOf()`, `StringBuilder.replace()`
- Input: `"abababab"`, `"ab"`, `"cd"`
- Output: `"cdcdcdcd"`

9. Find and Replace a Substring

- **Question**: Write a function to find the first occurrence of a substring and replace it with another string using `StringBuilder`.
- **Method(s) used**: `StringBuilder.indexOf()`, `StringBuilder.replace()`
- Input: `"hello world"`, `"world"`, `"Java"`
- Output: `"hello Java"`

10. String Compression

- **Question**: Implement basic string compression using counts of repeated characters. For example, `"aaabbbccc"` becomes `"a3b3c3"`.
- **Method(s) used**: `StringBuilder.append()`
- Input: `"aaabbbccc"`
- Output: `"a3b3c3"`

11. Palindrome Substring Construction

- **Question**: Given a string, find the longest palindromic substring and build it using `StringBuilder`.
- **Method(s) used**: `StringBuilder.append()`, `StringBuilder.reverse()`
- Input: `"babad"`
- Output: `"bab"` (or `"aba"`)

12. Remove a Substring from a String

- **Question**: Write a function that removes all occurrences of a given substring from a string using `StringBuilder`.
- **Method(s) used**: `StringBuilder.indexOf()`, `StringBuilder.delete()`
- Input: `"aabbcc"`, `"bb"`
- Output: `"aacc"`

13. Concatenate Multiple Strings Efficiently

- **Question**: Given an array of strings, concatenate them into one string efficiently using `StringBuilder` (instead of using `+`).
- **Method(s) used**: `StringBuilder.append()`
- Input: `["Java", " is", " awesome"]`
- Output: `"Java is awesome"`

14. Generate a Repeated String

- **Question**: Write a function to generate a string where a given string is repeated `n` times using `StringBuilder`.
- **Method(s) used**: `StringBuilder.append()`
- Input: `"abc"`, `3`
- Output: `"abcabcabc"`

15. Reverse Words in a Sentence

- **Question**: Write a function to reverse the words in a given sentence (without reversing the characters of each word) using `StringBuilder`.
- **Method(s) used**: `StringBuilder.append()`, `String.split()`
- Input: `"Hello World"`
- Output: `"World Hello"`

16. Remove All Vowels

- **Question**: Write a function to remove all vowels from a string using `StringBuilder`.
- **Method(s) used**: `StringBuilder.append()`, loop to check for vowels
- Input: `"hello world"`
- Output: `"hll wrld"`

17. Insert String at the Beginning or End

- **Question**: Write a function to insert a given string at the beginning or end of another string using `StringBuilder`.
- **Method(s) used**: `StringBuilder.insert()`, `StringBuilder.append()`
- Input: `"world"`, `"Hello "`, position: "start"
- Output: `"Hello world"`

18. Count and Replace Consecutive Repeated Characters

- **Question**: Write a function to count consecutive repeated characters and replace them with the count, such as `"aaabbcccc" → "a3b2c4"`.
- **Method(s) used**: `StringBuilder.append()`
- Input: `"aaabbcccc"`
- Output: `"a3b2c4"`

19. Create a String from a Character Array

- **Question**: Given a character array, write a function to build a string from the array using `StringBuilder`.
- **Method(s) used**: `StringBuilder.append()`
- Input: `['H', 'e', 'l', 'l', 'o']`
- Output: `"Hello"`

20. Generate a String Pattern

- **Question**: Write a function that generates a string pattern based on a number `n`, such as `n = 4` producing the pattern `"1234, 123, 12, 1"`.
- **Method(s) used**: `StringBuilder.append()`, loop

- Input: `4`
- Output: `"1234, 123, 12, 1"`

These questions test the candidate's ability to efficiently manipulate strings using `StringBuilder` (or `StringBuffer`), which is ideal for modifying and appending to strings due to its mutable nature. It's a common technique used to avoid the performance penalty that arises from using immutable `String` objects for multiple string modifications.